Module 2: A+ Peripherals and Power Supply

DAY 5:-

Task 1:

	Ethernet Cables	Fiber Optic Cables
Signal type	Electrical signals over copper wires	Light pulses over glass/ plastic strands
Max distance	100 meters before signal degrades	Tens of kilometers without a repeater
Cost & installation	Cheap, easy to terminate and install	Expensive, needs specialized tools
Real-world example	home Wi-Fi router connecting to a desktop PC or smart TV	college campus backbone linking buildings across a large campus where distance and bandwidth needs exceed what copper can handle

Task 2:

Straight through Ethernet Cable

Task 3:

Attaching an RJ45 Connector to a CAT6 Cable

1. Strip the outer jacket
Use the crimper's built-in stripper to remove about 1.5 inches of the outer plastic jacket from the end of the CAT6 cable, exposing the 4 twisted wire pairs inside.

2. **Untwist and arrange the wire**
Use the crimper's built-in stripper to remove about 1.5 inches of the outer plastic jacket from the end of the CAT6 cable, exposing the 4 twisted wire pairs inside.

3. **Trim the wires evenly**
Line up all 8 wires flat and tight side by side, then trim them with the crimper's cutting blade so they're all the same length - about half an inch from the jacket.

4. **Insert wires into the RJ45 connector**
Slide the flat, ordered wires into the RJ45 connector, making sure each wire reaches all the way to the front of the connector and stays in the correct pin order. The jacket should also slide slightly into the connector for the strain relief.

5. **Crimp the connector**
Insert the loaded connector into the crimping tool's RJ45 slot and squeeze firmly. This pushes the gold pins down through the insulation to make contact with each wire and locks the plastic tab that grips the cable jacket.

6. **Repeat on the other end**
Do the same process on the opposite end of the cable, again using T568B order, since this makes it a straight-through cable suitable for connecting a PC to a switch or router.

7. **Test the cable**
Plug both ends into a cable tester to confirm all 8 wires are making proper contact and there are no shorts or open connections.

Task 4:

Gaming Café Setup — 10 PCs for LAN Gaming

For a 10-PC gaming café built for **LAN gaming**, I'd use **Ethernet (CAT6) cables** with an **RJ45 crimper**, not fiber. Since all 10 PCs are in the same room or building, the distance is well within Ethernet's 100m limit, so fiber's long-distance advantage isn't needed here — CAT6 already supports up to 10 Gbps, more than enough for LAN gaming's low-latency requirements. Fiber and a cleaver would add unnecessary cost and complexity (specialized tools, splicing skill, pricier cable) for a setup where budget matters and the distances are short. Ethernet with a switch feeding all 10 PCs gives fast, low-latency, budget-friendly wired connections — exactly what competitive LAN gaming needs.